

Technology Stack

Date	15 March 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 Templates	Jinja2 templates are served directly by Flask, allowing dynamic rendering of the prediction form and result page without a separate frontend framework. HTML5/CSS3 ensures a clean, responsive layout with no extra dependencies.
2	Backend / Server-Side	Python 3, Flask	Flask is a lightweight Python web framework that integrates naturally with scikit-learn and joblib. Python was chosen because all data preprocessing, model training, and prediction logic are written in Python, keeping the stack consistent and simple.
3	Machine Learning / Core Logic	scikit-learn, pandas, numpy, joblib	scikit-learn provides the classification algorithms used to train and predict HDI categories. pandas and numpy handle data loading and feature preprocessing. joblib serialises the trained model to model.pkl for fast loading at runtime.
4	Database / Data Storage	Flat file (CSV) + model.pkl	The HDI indicator dataset is distributed as CSV files from public sources (UNDP / World Bank). No relational database is needed; the trained model is stored as a serialised file (model.pkl) and loaded on app startup, keeping infrastructure minimal.
5	Cloud / Hosting / Deployment	Render / Heroku (Gunicorn + Nginx)	Render and Heroku both support Python/Flask apps with free-tier options suitable for a student project. Gunicorn acts as the production WSGI server and Nginx as the reverse proxy, matching the architecture defined in the solution design.

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6	Version Control & CI/CD	Git, GitHub	GitHub hosts the project repository and enables team collaboration through branches and pull requests. GitHub Actions can automate testing and deployment to the hosting platform, providing a basic CI/CD pipeline at no cost.